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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

NON FERROUS EXTRACTIVE METALLURGY

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	i.	List six major non-ferrous metals which are commercially extractable from ores available in India.	03
		ii.	Draw a simple flow chart for the production of high purity copper from the sulphide ores indicating the various unit operations involved.	02
		iii.	Why a "side-blown" converter is used and not a "bottom-blown" unit in the extraction of copper?	02
	b.	i.	Write a brief note on the Noranda process for the extraction of copper with a schematic sketch.	04
		ii.	What are the major steps involved in the refining of lead bullion.	03
2.	a.	i.	Describe the horizontal retort process for the production of zinc with a schematic diagram.	04
		ii.	Bring out the major differences in the extraction of copper, lead and nickel from the sulphide ores.	03
	b.	i.	$Cu_2S + O_2 \rightarrow 2Cu + SO_2$ equilibrium constant K_1 $CuS + O_2 \rightarrow CuO + SO_2$ equilibrium constant K_2	02
			Derive the relation between $\log P_{O_2}$ and $\log P_{SO_2}$ for each of the reactions above.	03
		ii.	Write a brief note on the Imperial Smelting process for the production of zinc and lead.	02
		iii.	Why liquid lead is used for quenching zinc vapour at the top of the Imperial Smelting Process?	
3.	a.	i.	Write a brief note on the Beecher process for the production of TiO_2 from Ilmenite ore.	03
		ii.	Write a brief note on the Kroll's process for the extraction of titanium.	03
		iii.	Why carbon is used during the chlorination of TiO_2 to produce $TiCl_4$?	01
	b.	i.	Write any five differences between the Kroll's process with the Hunter's process in a Tabular form.	05
		ii.	What is the basic principle of operation of the Van-Arkel Iodide process? Write the chemical reactions.	02

OR

4. a. i. What is the principle of operation of the following processes: **05**

   i. Solvent extraction,

   ii. Ion exchange. Give one example in each case.

   ii. What are the major steps involved in the extraction of gold from secondary sources? **02**

   b. i. Compare Pyrometallurgy with Hydrometallurgy using a Tabular Form. **05**

   ii. What are the various unit processes involved in Hydrometallurgical operations? **02**

5. a. Calculate the partial pressure of magnesium vapour at equilibrium in the following reaction: $MgO(s) + Si(l) \rightarrow Mg(v)\ SiO_2(s)$ at T = 1200 °C, 1300 °C. State the assumptions involved. **04**

   What is the partial pressure of magnesium at T=1550 °C when the activity of MgO is unity and that of silicon is 0.6? **03**

   $G^0_{MgO(s)} = -544.7 - 0.104\ T$ kJ

   $G^0_{Si(l)} = 90.52 - 0.094\ T$ kJ

   $G^0_{Mg(v)} = 177.8 - 0.185\ T$ kJ

   $G^0_{SiO2(s)} = -838.0 - 0.140\ T$ kJ

   b. Describe the Pidgeon process of the production of magnesium with a sketch of retort; chemical reaction; temperature of reaction; vacuum level applied. **07**

OR

6. a. Describe the Hall-Heroult process for the production of aluminium from Al_2O_3. **07**

   b. i. Why a cryolite bath is used in the electrolysis of alumina? Illustrate with a simple sketch of the relevant phase diagram. **04**

   ii. Why the content of alumina in the bath is maintained at about 8% though the bath can dissolve up to 10% Al_2O_3 at the eutectic? **03**

7. a. Calculate the cell voltage for the following reactions at 1000 °C: **07**

   $Al_2O_3(s) \rightarrow 2Al(l) + 1.5O_2\ (g)$

   $Al_2O_3 + 3C \rightarrow 2Al(l) + 3CO(g)$

   $G^0_{Al2O3(l)} = -1540-0.2371T$ kJ

   $G^0_{C(s)} = 23.839-0.0244T$ kJ

   $G^0_{CO(g)} = -85.239-0.2395T$ kJ

   $G^0_{O2(g)} = 41.304-0.2589\ T$ kJ

   $G^0_{Al(l)} = 45.311-0.08603T$ kJ

   b. i. What are light and heavy Rare Earth metals? **02**

   ii. Mention any five applications of Rare Earths? **03**

   iii. Why is it very difficult to separate the Rare Earth elements? **02**

OR

8. **a.** Describe how bio-hydro metallurgy is used in the extraction of : (a) Copper, (b) Uranium. **07**

b. Describe the following processes with respect to the production of Rare Earth metals: **07**

i. Treatment of monazite ore (a summary flow chart may be used);

ii. Liquid-liquid separation